## Planetary Motion (Supplementary Problems)

**1.** Galileo measured the orbital sizes of Jupiter's moons using the diameter of Jupiter as a unit of measure. He found that Io, the closest moon to Jupiter, had a period of 1.8 days and was 4.2 units from the center of Jupiter. Callisto, the fourth moon from Jupiter, had a period of 16.7 days. Using the same units that Galileo used, predict Callisto's distance from Jupiter.

**2.** If Ganymede, one of Jupiter's moons, has a period of 32 days, how many units are there in its orbital radius? Use the information given in Example Problem 1.

**3.** An asteroid revolves around the Sun with a mean orbital radius twice that of Earth's. Predict the period of the asteroid in Earth years.

**4.** From Table 7-1, on page 173, you can find that, on average, Mars is 1.52 times as far from the Sun as Earth is. Predict the time required for Mars to orbit the Sun in Earth days.

**5.** The Moon has a period of 27.3 days and a mean distance of $3.90\times10^5$ km from the center of Earth. a. Use Kepler's laws to find the period of a satellite in orbit $6.70\times10^3$ km from the center of Earth. b. How far above Earth's surface is this satellite?

**6.** Using the data in the previous problem for the period and radius of revolution of the Moon, predict what the mean distance from Earth's center would be for an artificial satellite that has a period of exactly 1.00 day.

**7.** Assume that a satellite orbits Earth 225 km above its surface. Given that the mass of Earth is $5.97\times10^{24}$ kg and the radius of Earth is $6.38\times10^{6}$ m, what are the satellite's orbital speed and period?

**8.** **Uranus** Uranus requires 84 years to circle the Sun. Find Uranus's orbital radius as a multiple of Earth's orbital radius.

**9.** Jupiter is 5.2 times farther from the Sun than Earth is. Find Jupiter's orbital period in Earth years.

**10.** **Venus** Venus has a period of revolution of 225 Earth days. Find the distance between the Sun and Venus as a multiple of Earth's orbital radius.

**11.** If a small planet, D, were located 8.0 times as far from the Sun as Earth is, how many years would it take the planet to orbit the Sun?

**12.** Assume that a satellite orbits Earth 225 km above its surface. Given that the mass of Earth is $5.97\times10^{24}$ kg and the radius of Earth is $6.38\times10^{6}$ m, what are the satellite's orbital speed and period?

**13.** Suppose that the satellite in Example Problem 12 is moved to an orbit that is 24 km larger in radius than its previous orbit. What would its speed be? Is this faster or slower than its previous speed?

**14.** Use Newton's thought experiment on the motion of satellites to solve the following.

a. Calculate the speed that a satellite shot from a cannon must have to orbit Earth 150 km above its surface.

b. How long, in seconds and minutes, would it take for the satellite to complete one orbit and return to the cannon?

**15.** Use the data for Mercury in Table 7-1 on page 173 to find the following.
a. the speed of a satellite that is in orbit 260 km above Mercury's surface
b. the period of the satellite

**16.** Two spheres are placed so that their centers are 2.6 m apart. The force between the two spheres is $2.75\times10^{-12}$ N. What is the mass of each sphere if one sphere is twice the mass of the other sphere?

**17.** The Moon is $3.9\times10^5$ km from Earth's center and $1.5\times10^8$ km from the Sun's center. If the masses of the Moon, Earth, and the Sun are $7.3\times10^{22}$ kg, $6.0\times10^{24}$ kg, and $2.0\times10^{30}$ kg, respectively, find the ratio of the gravitational forces exerted by Earth and the Sun on the Moon.

**18.** **Figure 7-22** shows a Cavendish apparatus like the one used to find *G*. It has a large lead sphere that is 5.9 kg in mass and a small one with a mass of 0.047 kg. Their centers are separated by 0.055 m. Find the force of attraction between them.

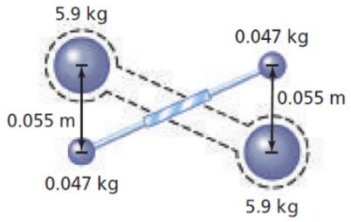



Figure 7-22

**19.** Tom has a mass of 70.0 kg and Sally has a mass of 50.0 kg. Tom and Sally are standing 20.0 m apart on the dance floor. Sally looks up and sees Tom. She feels an attraction. If the attraction is gravitational, find its size. Assume that both Tom and Sally can be replaced by spherical masses.

**20.** Two balls have their centers 2.0 m apart, as shown in **Figure 7-23.** One ball has a mass of 8.0 kg. The other has a mass of 6.0 kg. What is the gravitational force between them?

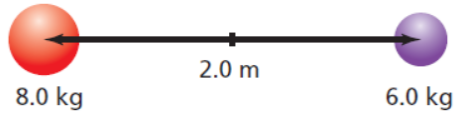



Figure 7-23

**21.** Two bowling balls each have a mass of 6.8 kg. They are located next to each other with their centers 21.8 cm apart. What gravitational force do they exert on each other?

**22.** Assume that you have a mass of 50.0 kg. Earth has a mass of $5.97\times10^{24}$ kg and a radius of $6.38\times10^{6}$ m. a. What is the force of gravitational attraction between you and Earth? b. What is your weight?

**23.** The gravitational force between two electrons that are 1.00 m apart is $5.54\times10^{-71}$ N. Find the mass of an electron.

**24.** A 1.0-kg mass weighs 9.8 N on Earth's surface, and the radius of Earth is roughly $6.4\times10^{6}$ m. a. Calculate the mass of Earth. b. Calculate the average density of Earth.

**25.** Mimas, one of Saturn's moons, has an orbital radius of $1.87\times10^{8}$ m and an orbital period of about 23.0 h. Use Newton's version of Kepler's third law to find Saturn's mass.

**26.** The Moon is $3.9\times10^{8}$ m away from Earth and has a period of 27.33 days. Use Newton's version of Kepler's third law to find the mass of Earth. Compare this mass to the mass found in problem 13.

**27.** Halley's Comet Every 74 years, comet Halley is visible from Earth. Find the average distance of the comet from the Sun in astronomical units (AU).

**28.** Area is measured in $m^2$, so the rate at which area is swept out by a planet or satellite is measured in $m^2/s$. a. How quickly is an area swept out by Earth in its orbit about the Sun? b. How quickly is an area swept out by the Moon in its orbit about Earth? Use $3.9\times10^8$ m as the average distance between Earth and the Moon, and 27.33 days as the period of the Moon.

**29.** Satellite A geosynchronous satellite is one that appears to remain over one spot on Earth, as shown in Figure 7-24. Assume that a geosynchronous satellite has an orbital radius of $4.23\times10^7$ m. a. Calculate its speed in orbit. b. Calculate its period.

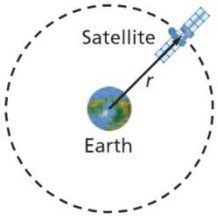



**Figure 7-24** (Not to scale)

**30. Asteroid** The asteroid Ceres has a mass of $7\times10^{20}$ kg and a radius of 500 km.

**a.** What is *g* on the surface of Ceres?

**b.** How much would a 90-kg astronaut weigh on Ceres?

**31.** A 1.25-kg book in space has a weight of 8.35 N. What is the value of the gravitational field at that location?

**32.** The Moon's mass is $7.34\times10^{22}$ kg, and it is $3.8\times10^{8}$ m away from Earth. Earth's mass is $5.97\times10^{24}$ kg.
**a.** Calculate the gravitational force of attraction between Earth and the Moon.
**b.** Find Earth's gravitational field at the Moon.

**33.** Two 1.00-kg masses have their centers 1.00 m apart. What is the force of attraction between them?

**34.** The radius of Earth is about $6.38\times10^3$ km. A $7.20\times10^3$-N spacecraft travels away from Earth. What is the weight of the spacecraft at the following distances from Earth's surface?

**a.** $6.38\times10^3$ km

**b.** $1.28\times10^4$ km

**35. Rocket** How high does a rocket have to go above Earth's surface before its weight is half of what it is on Earth?

**36.** Two satellites of equal mass are put into orbit 30.0 m apart. The gravitational force between them is $2.0\times10^{7}$ N.

**a.** What is the mass of each satellite?

**b.** What is the initial acceleration given to each satellite by gravitational force?

**37.** Two large spheres are suspended close to each other. Their centers are 4.0 m apart, as shown in **Figure 7-25.** One sphere weighs $9.8\times10^2$ N. The other sphere has a weight of $1.96\times10^2$ N. What is the gravitational force between them?

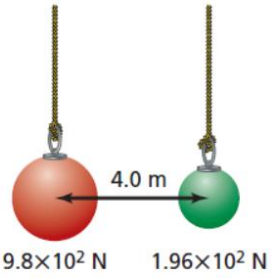



Figure 7-25

**38.** Suppose the centers of Earth and the Moon are $3.9\times10^8$ m apart, and the gravitational force between them is about $1.9\times10^{20}$ N. What is the approximate mass of the Moon?

**39.** On the surface of the Moon, a 91.0-kg physics teacher weighs only 145.6 N. What is the value of the Moon's gravitational field at its surface?

**40.** The mass of an electron is $9.1\times10^{-31}$ kg. The mass of a proton is $1.7\times10^{-27}$ kg. An electron and a proton are about $0.59\times10^{10}$ m apart in a hydrogen atom. What gravitational force exists between the proton and the electron of a hydrogen atom?

**41.** Consider two spherical 8.0-kg objects that are 5.0 m apart.

**a.** What is the gravitational force between the two objects?

**b.** What is the gravitational force between them when they are $5.0\times10^{1}$ m apart?

**42.** If you weigh 637 N on Earth's surface, how much would you weigh on the planet Mars? Mars has a mass of $6.42\times10^{23}$ kg and a radius of $3.40\times10^{6}$ m.

**43.** Using Newton's version of Kepler's third law and information from Table 7-1 on page 173, calculate the period of Earth's Moon if the orbital radius were twice the actual value of $3.9\times10^{8}$ m.

**44.** Find the value of *g*, acceleration due to gravity, in the following situations.
**a.** Earth's mass is triple its actual value, but its radius remains the same.
**b.** Earth's radius is tripled, but its mass remains the same.
**c.** Both the mass and radius of Earth are doubled.

**45. Astronaut** What would be the strength of Earth's gravitational field at a point where an 80.0-kg astronaut would experience a 25.0 percent reduction in weight?

**46.** Use the information for Earth in Table 7-1 on page 173 to calculate the mass of the Sun, using Newton's version of Kepler's third law. Earth's gravitational field is 7.83 N/kg at the altitude of the space shuttle. At this altitude, what is the size of the force of attraction between a student with a mass of 45.0 kg and Earth?

**47.** Use the data from Table 7-1 on page 173 to find the speed and period of a satellite that orbits Mars 175 km above its surface.

**48.** **Satellite** A satellite is placed in orbit, as shown in **Figure 7-26,** with a radius that is half the radius of the Moon's orbit. Find the period of the satellite in units of the period of the Moon.

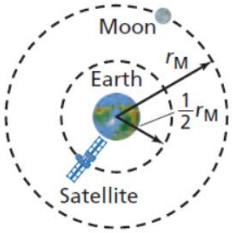



Figure 7-26

**49. Cannonball** The Moon's mass is $7.3\times10^{22}$ kg and its radius is 1785 km. If Newton's thought experiment of firing a cannonball from a high mountain were attempted on the Moon, how fast would the cannonball have to be fired? How long would it take the cannonball to return to the cannon?

**50.** The period of the Moon is one month. Answer the following questions assuming that the mass of Earth is doubled.

**a.** What would the period of the Moon be? Express your results in months.

**b.** Where would a satellite with an orbital period of one month be located?

**c.** How would the length of a year on Earth be affected?

**51.** How fast would a planet of Earth's mass and size have to spin so that an object at the equator would be weightless? Give the period of rotation of the planet in minutes.

**52.** **Car Races** Suppose that a Martian base has been established and car races are being considered. A flat, circular race track has been built for the race. If a car can achieve speeds of up to 12 m/s, what is the smallest radius of a track for which the coefficient of friction is 0.50?

**53.** ***Apollo 11*** On July 19, 1969, *Apollo 11's* revolution around the Moon was adjusted to an average orbit of 111 km. The radius of the Moon is 1785 km, and the mass of the Moon is $7.3\times10^{22}$ kg.

**a.** How many minutes did *Apollo 11* take to orbit the Moon once?

**b.** At what velocity did *Apollo 11* orbit the Moon?